

Semiconductor wafers --- Solution

Y20 (2020) — English translation

A1 Obtain the values of I_{AB} , U_{CD} , I_{DA} and U_{BC} for 15 different connections to the plate. ??

A2 Compute $R_{AB,CD}$, $R_{BC,DA}$ and y for each of the connections. Round all values to the nearest 4 significant figures. 0.90

A3 The formula $\frac{y-1}{y+1} = \frac{1}{x} \operatorname{arch}\left(\frac{e^x}{2}\right)$ can be converted to the form $e^x = e^{\alpha x} + e^{\beta x}$. Find α and β . Express your answer using y . 0.10

A4 The equation $e^x = e^{\alpha x} + e^{\beta x}$ is transcendental and cannot be solved analytically. Solve it numerically, i.e. for each y , find e^x (to 4 significant figures). 3.00

$R_{AB,CD}$	$R_{BC,DA}$	y	e^x	x	$\frac{x}{y+1}$
$6,612 \cdot 10^5$	$3,805 \cdot 10^5$	1,738	2,037	0,7115	$2,599 \cdot 10^{-1}$
$9,335 \cdot 10^5$	$2,366 \cdot 10^5$	3,945	2,233	0,7989	$1,616 \cdot 10^{-1}$
$1,117 \cdot 10^6$	$1,793 \cdot 10^5$	6,231	2,425	0,8858	$1,225 \cdot 10^{-1}$
$1,469 \cdot 10^6$	$1,057 \cdot 10^5$	13,90	2,932	1,076	$7,221 \cdot 10^{-1}$
$1,793 \cdot 10^6$	$6,600 \cdot 10^4$	27,17	3,562	1,270	$4,508 \cdot 10^{-2}$
$1,917 \cdot 10^6$	$5,545 \cdot 10^4$	34,57	3,846	1,347	$3,787 \cdot 10^{-2}$
$2,099 \cdot 10^6$	$4,286 \cdot 10^4$	48,97	4,318	1,463	$2,928 \cdot 10^{-2}$
$2,175 \cdot 10^6$	$3,854 \cdot 10^4$	56,44	4,534	1,512	$2,632 \cdot 10^{-2}$
$2,301 \cdot 10^6$	$3,227 \cdot 10^4$	71,31	4,923	1,594	$2,204 \cdot 10^{-2}$
$2,580 \cdot 10^6$	$2,189 \cdot 10^4$	117,8	5,913	1,777	$1,496 \cdot 10^{-2}$
$2,975 \cdot 10^6$	$1,267 \cdot 10^4$	234,8	7,699	2,041	$8,656 \cdot 10^{-3}$
$3,278 \cdot 10^6$	$8,353 \cdot 10^3$	392,5	9,442	2,245	$5,705 \cdot 10^{-3}$
$3,493 \cdot 10^6$	$6,224 \cdot 10^3$	561,2	10,91	2,390	$4,251 \cdot 10^{-3}$
$3,689 \cdot 10^6$	$4,758 \cdot 10^3$	775,4	12,46	2,523	$3,250 \cdot 10^{-3}$
$3,808 \cdot 10^6$	$4,042 \cdot 10^3$	942,1	13,51	2,603	$2,760 \cdot 10^{-3}$

A5 Having plotted a graph in suitable coordinates, determine the numerical value ρ of the semiconductor. Round your answer to 3 significant figures. Take the thickness of the material equal to $d = 1$ mm. 2.00

$$\rho = (2.3 \cdot 10^3 \pm 5) \text{ OM}$$

B1 Remove dependency $n(T)$. ??

$\ln n$	$\frac{1}{T}, \text{K}^{-1}$	$\ln(nT^{\frac{-3}{4}})$	$\ln(nT^{\frac{-3}{2}})$
54,165	$3,33 \cdot 10^{-2}$	51,614	
54,745	$2,50 \cdot 10^{-2}$	51,978	
55,124	$2,00 \cdot 10^{-2}$	52,190	
55,405	$1,67 \cdot 10^{-2}$	52,335	
55,618	$1,43 \cdot 10^{-2}$	52,432	
55,796	$1,25 \cdot 10^{-2}$	52,510	
55,943	$1,11 \cdot 10^{-2}$	52,568	
56,068	10^{-2}	52,614	
56,176	$9,09 \cdot 10^{-3}$	52,651	
56,275	$8,33 \cdot 10^{-3}$	52,685	
56,357	$7,69 \cdot 10^{-3}$		
56,359	$4,98 \cdot 10^{-3}$		
56,359	$3,676 \cdot 10^{-3}$		
56,359	$2,915 \cdot 10^{-3}$		
56,359	$2,415 \cdot 10^{-3}$		
56,360	$1,590 \cdot 10^{-3}$		
58,666	$1,186 \cdot 10^{-3}$		48,561
58,724	$1,172 \cdot 10^{-3}$		48,601
58,782	$1,159 \cdot 10^{-3}$		48,642
58,839	$1,145 \cdot 10^{-3}$		48,681
58,896	$1,133 \cdot 10^{-3}$		48,721
58,951	$1,120 \cdot 10^{-3}$		48,760
59,005	$1,107 \cdot 10^{-3}$		48,797
59,056	$1,095 \cdot 10^{-3}$		48,834
59,111	$1,083 \cdot 10^{-3}$		48,869
59,162	$1,072 \cdot 10^{-3}$		48,905
59,213	$1,060 \cdot 10^{-3}$		48,939

B2) $T_i = (128 \pm 5)\text{K}$, $T_s = (780 \pm 40)\text{K}$

B3) $E_d = (1,18 \pm 0,02) \cdot 10^{-21} \text{ Дж}$

B4) $\Delta E_g = (8,23 \pm 0,11) \cdot 10^{-20} \text{ Дж}$



B2 Plot the dependence $\ln n \left(\frac{1}{T} \right)$. Approximate three sections of this graph with straight lines and determine the values of T_i and T_s . Estimate the error of the obtained values.

1.60

B3 Find E_d .

1.20

B4 Find ΔE_g .

1.20